

Full Stack Junior Application Task

Project: Credit Card Payment System

Frontend: React + Tailwind CSS

Backend: Django + FastAPI

Database: MySQL

MODULE 1: USER AUTHENTICATION (DJANGO)

- User Registration
- User Login using JWT Authentication
- Logout
- Password Encryption
- Protected Routes

MODULE 2: CARD MANAGEMENT (DJANGO)

- Add Credit/Debit Card
- View Saved Cards
- Delete Card
- Do NOT store actual card numbers
- Store only masked card and last 4 digits

MODULE 3: PAYMENT PROCESSING (FASTAPI)

- Make a Payment
- Simulate Payment Success/Failure
- Initial Status: PENDING
- Final Status: SUCCESS or FAILED

MODULE 4: TRANSACTION MANAGEMENT (DJANGO)

- View Transaction History
- Filter by Date, Amount, and Status
- Admin Export to CSV

MODULE 5: ADMIN PANEL (DJANGO)

- Manage Users
- View Cards
- View Transactions
- Daily Payment Summary

MODULE 6: FRONTEND (REACT + TAILWIND)

- Register Page
- Login Page
- Dashboard
- Add Card
- Make Payment
- Transaction History
- Admin Dashboard

MODULE 7: DATABASE (MYSQL)

- Users Table
- Cards Table
- Transactions Table
- Admin Logs

MODULE 8: SECURITY RULES

- No CVV storage
- Encrypted Passwords
- JWT Authentication
- Input Validation
- SQL Injection Protection

MODULE 9: API DOCUMENTATION

- FastAPI Swagger Enabled
- Django API Documentation
- Postman Collection

MODULE 10: DOCKER & DEPLOYMENT

- Dockerfile for Frontend, Django, FastAPI, MySQL
- docker-compose required

MODULE 11: TESTING REQUIREMENTS

- Unit Tests for Authentication
- Card Management Tests
- Payment Tests
- Minimum 50% Test Coverage

MODULE 12: GIT & DOCUMENTATION

- Proper Commit Messages
- README must include Setup, API Docs, DB Schema, Screenshots

■ Time Limit: 4 Days (Strict Deadline)

Evaluation Criteria: - Functionality: 40% - Code Quality: 20% - UI/UX: 15% - Security: 10% - Testing: 10% - Documentation: 5%

Final Submission Must Include: - GitHub Repository Link - Database Dump - Postman Collection - UI Screenshots - Admin Credentials